

ONYX & XELOR

End product — the production architecture

What the MVP becomes when it carries real factories: the managed service that replaces each development container, how a release ships without downtime, how the data and AI platforms are fed — and what we take, and deliberately do **not** take, from the platforms Agoda and MakeMyTrip actually run.

TARGET: AWS AP-SOUTH-1 PATTERNED ON AGODA · MAKEMYTRIP 20 AUG 2026

01 From MVP to production, layer by layer

Nothing in the application changes shape. Every row below swaps a single development container for the managed service that carries the same contract under load, with a blast radius and a recovery story.

LAYER	MVP TODAY	END PRODUCT
Compute	One Node process per app, started by hand	EKS across three availability zones — a deployment each for <code>onyx-api</code> , <code>xelor-api</code> , <code>web</code> and the workers, autoscaled on request rate and queue depth.
Database	One Postgres 17 container, one volume	Aurora PostgreSQL 17 — writer plus two readers across AZs, point-in-time recovery to 35 days. Shared schema and FORCE RLS are unchanged.
Cache & jobs	One Valkey container serving both	ElastiCache for Valkey in cluster mode, split endpoints so a cache flush can never drop a queued job.
Event spine	Outbox → BullMQ, in-process consumers	Outbox → BullMQ for work that must run; MSK (Kafka) for the fan-out that others read — integrations, analytics, telemetry, ML.
Identity	One Keycloak container, file realm	Keycloak cluster on its own Aurora database, one realm per environment, sessions replicated.
Files & PDFs	Local volume, one Gutenberg	S3 with versioning and SSE-KMS behind CloudFront; Gutenberg as an autoscaled render pool.
Retrieval	pgvector inside the app database	pgvector on its own instance class, moving to OpenSearch only when the copilot corpus outgrows it.
Models	Offline stub, or Ollama on a laptop	Router with provider failover, plus self-hosted vLLM on GPU nodes for anything that touches personal data.
Observability	Console logs	OpenTelemetry SDK → Collector → Prometheus and Grafana for metrics, Loki for logs, Tempo for traces, Sentry for errors.
Delivery	Build and restart	GitOps with Argo CD, immutable images, canary by tenant cohort, automatic rollback on business metrics.
Service mesh	Direct HTTP between two processes	Istio — mTLS between every service, retries and timeouts as policy rather than code, and the traffic shifting a canary needs.
Secrets	<code>.env</code> files on the machine	Secrets Manager with KMS keys and scheduled rotation; nothing sensitive in an image, a repository or a log line.
Environments	One laptop, one world	Development, staging and production as separate AWS accounts, so a mistake in one cannot reach the data in another.
Plant edge	In-process simulator	A per-plant edge agent over mTLS with store-and-forward and a durable command journal, so a dropped link delays work rather than losing it.

02 What we copy from them, and what we refuse to

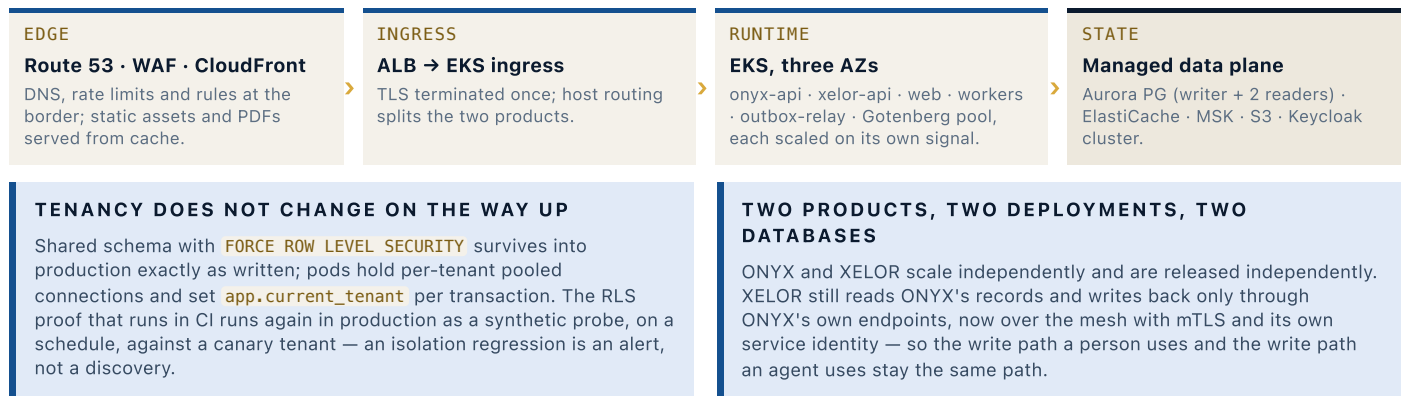
COPY · CANARY ON BUSINESS METRICS Agoda ships to one cluster first and watches booking rate, latency and exceptions before it goes to every datacentre. Our equivalent watches order-confirm rate, mission-approval latency and RLS denials on one tenant cohort — a release that moves any of them rolls itself back.	COPY · ONE EVENT SPINE, MANY READERS Agoda runs ~2,000 Kafka topics; MakeMyTrip pairs Kafka with Avro and RabbitMQ. The pattern we need is theirs: producers write once, and analytics, integrations and ML each read at their own pace. We feed it from the outbox we already have.	REFUSE · A PRIVATE CLOUD Agoda built Fleet, Buckbeak and its own DBaaS over ~1,200 database instances and 2,500 services. That is an answer written by a thousand engineers. Five engineers buy the managed equivalents — EKS, Aurora, MSK, ElastiCache — and spend the saved years on the factory.
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Sized for the right problem. Agoda moves 1.8–2 trillion events a day; MakeMyTrip serves 120k orders and 800k notifications a day at 44 ms p50. A manufacturing ERP for Indian MSMEs is a different shape of load — hundreds of plants, thousands of transactions each, every one of which must be correct, tenant-isolated and auditable eight years later. We take their *patterns* for delivery, eventing and observability, and we size the hardware for correctness rather than for consumer volume.

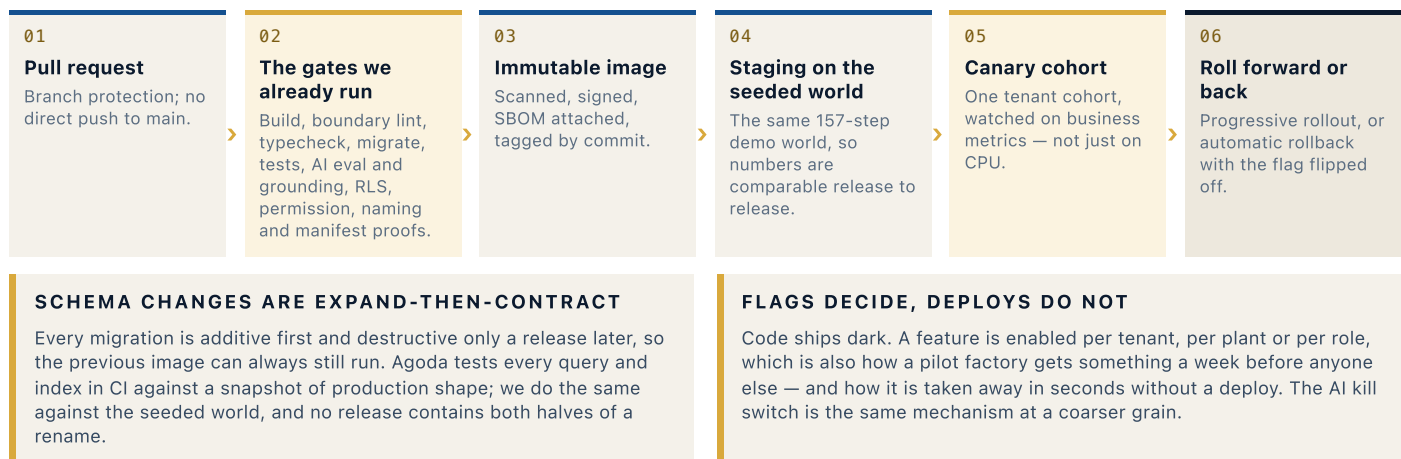
Researched from: Agoda Engineering on Kafka at 1.8T events/day and on moving 3M CI jobs to Kubernetes · The Pragmatic Engineer, “Inside Agoda’s Private Cloud” (Fleet, Buckbeak on KubeVela, Istio, Ceph, VAST, Iceberg, ~20PB, ~1,200 DB instances, ~2,500 services) · Databricks, “How MakeMyTrip achieved millisecond personalization” (Spark Structured Streaming Real-Time Mode, 44 ms p50, Aerospike, Redis, Kafka).

03 The production topology

One VPC in ap-south-1, three availability zones, two products that stay separate all the way down to their databases — because XELOR runs migrations ONYX has never seen.



04 How a release ships



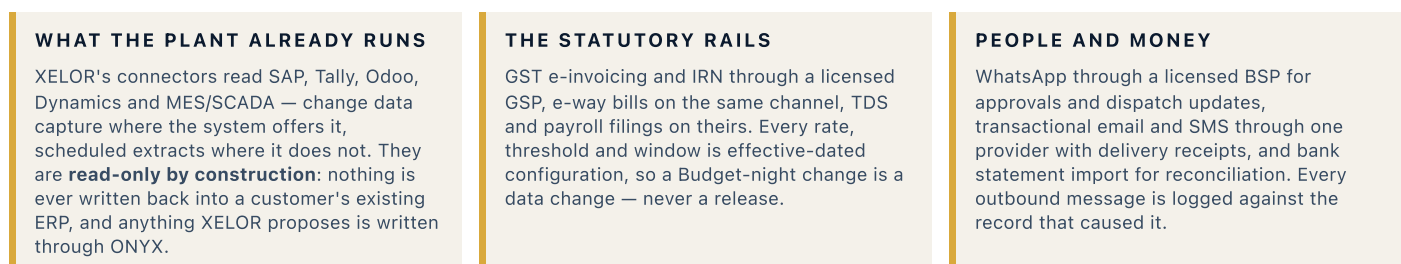
05 The data platform

One direction of flow, and no analytics query ever touching the transactional database.



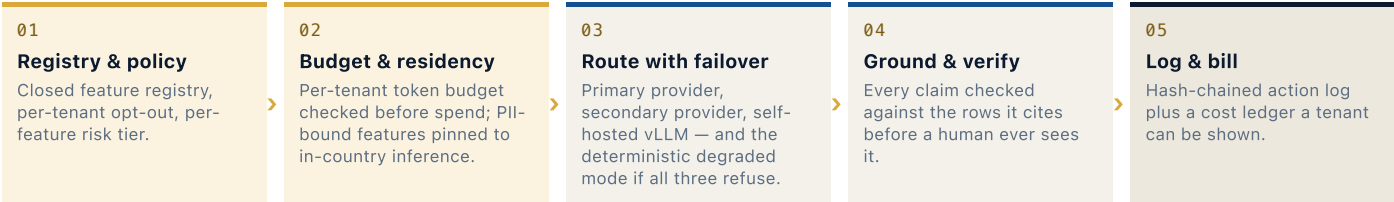
06 The integration surface

A production ERP is judged by what it connects to. Three surfaces, each with a named owner and a failure mode that is visible rather than silent.



07 The AI platform in production

The router the MVP already has becomes the control plane: the same four steps, now with money, residency and failure in them.



EVALUATION DOES NOT STOP AT THE MERGE

The golden-set and grounding gates keep blocking CI. Production adds shadow evaluation — a candidate model answers the same questions as the live one without reaching a user — plus drift alarms on refusal rate, grounding failures and cost per decision. A provider is promoted on evidence, not on a release note.

PERSONAL DATA DOES NOT LEAVE THE COUNTRY

DPDP is a routing rule, not a policy document: features that can carry personal data are pinned to self-hosted inference inside ap-south-1, and the router refuses to send them anywhere else. Egress is enforced where the call is made, so no module can opt out of it.

08 The agents at scale

DURABLE, NOT MERELY RETRIED

Node claims, heartbeats and bounded attempts already exist. Production runs them on a dedicated worker pool with per-tenant concurrency caps, so one plant's mission storm cannot starve another's, and any run can be replayed from its journal.

APPROVALS HAVE A CLOCK

A run waiting on a person is a run costing money. Approvals carry an SLA, escalate up the role chain, and reach the approver where they are — push, email, and WhatsApp through a licensed BSP.

THREE KILL SWITCHES

Per feature, per tenant, and global. Each one drops the affected feature to its declared degraded mode rather than to an error, so the factory keeps running while the model is off.

09 Reliability, security, and what the law requires

COMMITMENT	TARGET	HOW IT IS HELD
Availability	99.9% monthly, per product	Multi-AZ everywhere, PodDisruptionBudgets, error budgets that stop feature work when spent.
Latency	p95 read < 300 ms · p95 write < 800 ms	Read replicas for reporting, cursor pagination everywhere, no analytics on the writer.
Recovery	RPO 5 min · RTO 60 min	PITR to 35 days, cross-region backup copy, and a restore drill each quarter that is timed and recorded.
Secrets & keys	KMS · Secrets Manager · mTLS in mesh	No secret in an image or a repository; rotation scheduled, image scanning and SBOM on every build.
Tenant isolation	Proven, not asserted	The RLS and permission proofs run in CI and again as scheduled production probes against a canary tenant.
Indian statute	DPDP · CERT-In · MCA · GST	Data resident in ap-south-1, incident reporting inside the six-hour window, eight-year immutable audit retention on the hash chain, statutory rates as effective-dated config.
Assurance	SOC 2 Type II · ISO 27001 path	Controls mapped to the gates that already exist; annual third-party penetration test.

10 Capacity, cost and how it grows

THE UNIT THAT SCALES

Tenants share one cluster and one database until a plant's volume or its contract says otherwise; then it moves to its own Aurora instance without changing a line, because every query is already tenant-scoped. Nothing in the code knows which arrangement it is in.

WHERE THE MONEY GOES

Aurora and MSK dominate the bill long before compute does, and GPU inference dominates both if it is used carelessly — which is why the token budget sits in the router and the cost per decision is a monitored metric rather than a month-end surprise.

WHAT TRIGGERS THE NEXT STEP

Three signals, each with a number: RLS overhead past 20%, replica lag past the reporting SLA, or a single tenant past a quarter of cluster throughput. Any one of them opens a design review — the same rule the platform already applies in development.

11 Running it, and putting a plant on it

OPERATIONS

A severity ladder with a named on-call rota, a runbook per alert that has fired at least once, a public status page, and a blameless review inside five working days of any Sev-1. Support tiers are contractual: response times by plan, and a named engineer for pilot accounts.

PUTTING A PLANT LIVE

Tenant and plant provisioned from a template, masters imported from Tally or spreadsheets through the data-import module with a dry run before commit, roles mapped to the people who actually sign, then a fortnight running in parallel with the paper before cutover — and the AI features switched on last, one at a time.

One thing does not scale away. Every service above is replaceable — the model, the queue, the cloud. The invariant is not: **AI explains, it never decides.** Nothing reaches a customer, a supplier, a ledger or a machine without a person approving it, and every approval stays reconstructable — the cited rows, the model call, the permission held, and the hash chain proving none of it was edited since.